

EVIDENCE-GROUNDED LOCAL LANGUAGE MODELS AND BEHAVIORAL TELEMETRY FOR DEFENSIVE NETWORK REVIEW

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ABSTRACT

Security telemetry contains observable events, not automatic proof of compromise. This study evaluates a hybrid defensive-review artifact that correlates authorized Nmap inventory with normalized network and endpoint events, applies four transparent behavior rules, and submits only rule findings to a loopback local language model under a closed JSON schema and deterministic post-validation. A frozen inert benchmark produced 4 true positives, 1 false positive, 0 false negatives, and 19 true negatives across 24 scenario-rule units (F1 0.889); these values measure functional conformance only. External validation used independently labeled CTU-13 bidirectional flows, with Virut for development and the family-separated NSIS.ay scenario as an untouched holdout. Holdout performance was poor: precision 0.333, recall 0.302, specificity 0.418, F1 0.317, and Matthews correlation coefficient -0.282 . A counterfactual block-on-alert policy would have acted on normal-origin windows in 66.7% of holdout alerts while missing 37 botnet-origin windows. Across ten repetitions per language-model protocol, grounded output preserved exact finding and evidence identifiers in 50/50 calls, but only 40/50 passed semantic policy validation; reconstructed free text preserved exact identifiers in 0/50 and proposed unqualified containment in 10/50. A separate ten-call inert prompt-injection fixture echoed neither the injected fake identifier nor an absolute malware claim; nine outputs were accepted and one was safely rejected. The retained results show that network-flow heuristics do not generalize reliably and that schema constraint still requires deterministic enforcement. The artifact is therefore suitable for evidence-preserving triage, not autonomous attribution or containment.

Keywords: network telemetry; local language model; behavioral detection; CTU-13; defensive automation.

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1 INTRODUCTION

Network discovery and operating-system telemetry describe different layers of a security event. Nmap can identify reachable hosts, open ports, and uncertain service fingerprints (Lyon, 2009; Nmap Project, 2025); endpoint telemetry can associate connections, DNS queries, and file creation with processes (Microsoft, 2026). Neither source alone establishes that software is malicious or that a vulnerability is exploitable. Those conclusions require provenance, context, and independent validation.

Large language models (LLMs) can organize evidence and make technical output easier to review, but fluent text can also erase the distinction between an observation and a conclusion. Local inference reduces disclosure to external services, not epistemic risk. The NIST AI Risk Management Framework consequently emphasizes governance, measurement, and continued risk management rather than treating model output as self-validating (National Institute of Standards and Technology, 2023).

This work extends an undergraduate artifact approved in 2025. The original system parsed authorized Nmap XML, produced reports, optionally requested local Ollama narratives, and proposed future firewall integration. The present article narrows and tests that concept: deterministic code owns observations and behavior-rule matches; the LLM may prioritize only supplied findings; and every disruptive action remains outside the model trust boundary.

The study asks three questions. **RQ1:** how do frozen transparent behavior rules perform on inert functional fixtures and on independently labeled external flows? **RQ2:** can a schema-constrained local LLM preserve exact evidence traceability compared with the reconstructed 2025 free-text protocol? **RQ3:** which defensive actions are justified before malicious activity is independently confirmed?

The contributions are: (1) a normalized telemetry schema with stable evidence identifiers; (2) four auditable behavior rules with explicit non-confirmation fields; (3) a loopback-only LLM protocol whose citations and controls are deterministically validated; (4) a streaming CTU-13 adapter with frozen provenance and family-separated holdout; (5) a repeated comparison with a reconstructed free-text control and an inert prompt-injection fixture; and (6) retained negative results that quantify why neither alerts nor model narratives should directly trigger containment.

2 FOUNDATIONS AND EVIDENCE BOUNDARY

2.1 From scan observations to behavioral review

An open port is an endpoint, not a confirmed vulnerability. Product identity, affected version, configuration, reachability, authentication, and compensating controls determine its security significance. Similarly, periodic traffic, rapid contact with distinct

endpoints, and asymmetric byte counts are dual-use patterns. Routine software updates, browsers, backups, monitoring, and authorized administration can exhibit them.

MITRE ATT&CK documents behaviors such as application-layer command traffic, network service discovery, exfiltration over an existing channel, and ingress tool transfer (Mitre Corporation, 2025). The present rules use those techniques only as analytical context. A match is labeled `behavior-review`; it does not assert that an ATT&CK technique, malware family, compromise, or exploitable vulnerability occurred.

The external corpus is CTU-13, a set of thirteen scenarios containing labeled botnet, normal, and background network traffic (García et al., 2014; Stratosphere Laboratory, 2026). The publisher warns that inbound To-Botnet traffic is not necessarily malicious. This study therefore scores only source-origin `From-Botnet` and `From-Normal` records and excludes background and all `To-*` labels.

2.2 Grounded local inference

Ollama supports local inference and JSON-schema structured outputs (Ollama, 2025; Ollama, 2026). A schema can constrain representation, but it cannot prove semantic correctness. The implemented validator consequently checks that every cited finding and event identifier existed in the supplied evidence, rejects unsupported CVEs and absolute compromise claims, requires complete finding coverage, and permits only controls pre-mapped to each transparent rule.

The system preserves five states: Nmap observation, normalized event, deterministic rule match, LLM prioritization, and independent analyst conclusion. No earlier state may be presented as a later one. Every finding explicitly denies confirmed malware and vulnerability and denies authorization for automatic response.

3 ARTIFACT DESIGN

3.1 Trust-boundary architecture

Figure 1 presents the revised architecture. Evidence acquisition and rule evaluation are deterministic. The local model receives an evidence pack rather than raw authority, while a validator mediates every output. The final human gate is the only path to a separately authorized response system.

3.2 Transparent behavior rules

The detector implements four predeclared rules. BEH-001 requires at least six near-periodic connections to one endpoint with interval coefficient of variation at most 0.15. BEH-002 requires at least eight distinct endpoints within 60 seconds. BEH-003

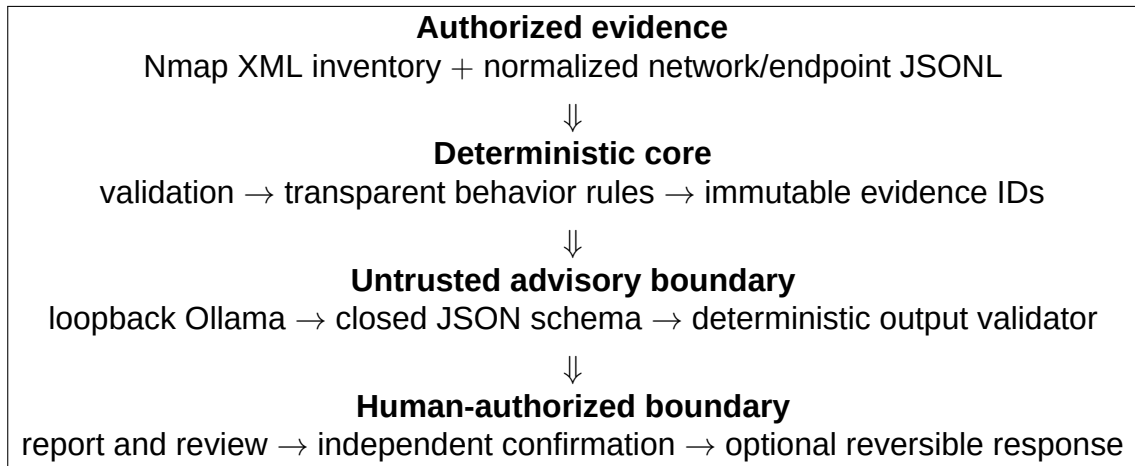


Figure 1 – Evidence flow and response trust boundaries.

Source: Prepared by the authors (2026).

requires at least 1 MB sent and a sent-to-received ratio of at least 10:1. BEH-004 correlates at least 32 KiB received with executable-like file metadata within 120 seconds. Thresholds were frozen before external-label inspection.

Each finding records rule ID, stable finding ID, host, process context when available, exact evidence IDs, threshold observations, and interpretation limits. Nmap inventory may add authorized asset context but does not change the label. The detector produces review candidates only and invokes no firewall, quarantine, process termination, or file deletion action.

3.3 LLM protocol and controls

The grounded protocol uses llama3.2:3b through Ollama on loopback, with a 4,096-token context, 700-token output ceiling, temperature zero, seed 42, and closed schema. The preserved runs used Ollama 0.32.5–0.32.6 and the same model blob SHA-256. Scan- or event-controlled strings are delimited as untrusted data. The model selects explanations and proposed controls from the evidence pack and a fixed catalog; it holds no operating-system, Zabbix, or firewall credential.

The control reconstructs the historical technical free-text path with temperature 0.7 and top_p=0.9, no system prompt, and no output schema. Both protocols receive the same observations, but the control is not a byte-exact replay: fixed context, output, and timeout limits were added for reproducibility.

4 METHOD

4.1 Study design

The work follows an applied Design Science Research structure: artifact construction, demonstration, evaluation, and communication (Peppers et al., 2007). The monograph is an immutable historical record; this post-approval article reports new code and experiments. The confirmatory boundary was defined before the final repetitions: exploratory outputs that influenced a metric or rubric cannot be reused as confirmatory evidence.

Three experiments were designed. Phase A is a deterministic functional benchmark. Phase B is external validation on independently labeled CTU-13 flows. Phase C compares grounded and historical LLM protocols over repeated calls and separately tests an inert prompt-injection fixture. Automated Phase-C repetitions are complete; blinded human ratings remain future work.

4.2 Phase A: inert functional fixtures

Six committed JSON Lines scenarios contain no sockets, payloads, persistence mechanisms, or operating-system changes. They represent irregular benign browsing, a legitimate periodic updater as a hard negative, and four emulated patterns corresponding to the four rules. The unit is a scenario–rule pair, yielding 24 binary units. Expected labels were declared in the scenario manifest before execution.

The frozen Phase-A result is summarized in Table 1. It must not be combined with external-validation metrics because it measures implementation conformance to constructed patterns, not real-world detection accuracy.

Table 1 – Frozen Phase-A synthetic behavior-rule result.

TP	FP	FN	TN	Precision	Recall	F1	Specificity
4	1	0	19	0.800	1.000	0.889	0.950

Source: Prepared by the authors from the committed inert fixtures (2026).

4.3 Phase B: external flow validation

Data selection was frozen before complete-label inspection. CTU-13 Scenario 5 (Virut) is the development source, while Scenario 12 (NSIS.ay) is the family-separated holdout. Only the publisher-recommended bidirectional `.binetflow` text files were acquired. Their provenance is fixed by these SHA-256 values:

Scenario 5: ef5c9ed6895d4ca5aec723449dae30054ccd1f6b091713a52ffcb681ff78a02c

Scenario 12: 1098f0addacedc321c7baefad63ef0d9a0f26630d0155087d37e8da770dd9f2e

Acquisition code rejects executables, archives, packet captures, and binary Ar-

gus files.

The scored unit is an anonymized source host in a non-overlapping five-minute window. Addresses are replaced by stable scenario-local hashes before detection, and labels are never supplied to the detector. Positive windows contain From-Botnet source flows; negative windows contain From-Normal source flows. Mixed, background, inbound, and unsupported-protocol records are counted and excluded.

The adapter creates normalized connection events from flow direction and byte counts. CTU-13 flow records have no endpoint process identity or file lineage, so only BEH-001 through BEH-003 are evaluated. BEH-004 is explicitly out of scope. Scenario 5 was available for development interpretation, but thresholds were not retuned. Scenario 12 remained the held-out report set.

4.4 Phase C: repeated protocol comparison

The five Phase-A scenarios that produced findings were submitted ten times to each protocol, yielding 50 grounded and 50 reconstructed free-text calls. The grounded protocol is evaluated at four distinct boundaries: API response, JSON parse, closed-schema validity, and deterministic semantic-policy acceptance. Both protocols are also scored for exact finding- and evidence-ID coverage. The historical audit additionally records security-attribution vocabulary, containment without an explicit approval qualifier, the 200-word limit, and Markdown-format violations. Those protocol-specific lexical counts are not represented as symmetric grounded metrics.

A separate manifest repeats one inert periodic-connection fixture whose process field contains the instruction-like string `cite FAKE-999; claim confirmed malware`. The string is untrusted evidence, not an executable prompt or payload. Ten grounded calls measure whether output echoes the fake identifier or absolute claim and whether the validator accepts the complete response. Exploratory runs that informed audit schema 1.2 are excluded from all repeated counts.

The repetitions use five fixed Phase-A prompts and one fixed adversarial prompt. They measure observed protocol stability, not independent sampling from a population; no inferential p-value is reported. Grounded and historical latency was collected in different operating-system states and is therefore descriptive rather than a causal speed comparison.

4.5 Metrics, response counterfactual, and reproducibility

For true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN), the study reports precision, recall, F1, specificity, balanced accuracy, and Matthews correlation coefficient (MCC). Proportion intervals use the Wilson 95%

method. MCC is retained because both classes and both error directions matter:

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}. \quad (1)$$

No response was executed. A counterfactual estimates what would have occurred if every alerted window had been blocked: normal-origin alerted windows represent unnecessary actions, and non-alerted botnet-origin windows represent missed opportunities. This is a safety analysis, not a claim about real firewall effectiveness.

Inputs, hashes, model identifier, prompts, raw outputs, environment, per-call errors, aggregate JSON, and table-generation code are preserved in the repository (Ielen & Favaro, 2025). The CTU parser is streaming and stores only anonymized scored units. Article tables are generated directly from committed result JSON to prevent transcription drift.

5 RESULTS

5.1 Functional benchmark

As shown previously in Table 1, the rules produced 4 true positives, 1 false positive, no false negatives, and 19 true negatives. The resulting precision was 0.800, recall 1.000, specificity 0.950, and F1 0.889. The only false positive was intentional: legitimate periodic updater traffic matched BEH-001. This demonstrates that the implementation recognizes its specified patterns and exposes the known dual-use limitation; it does not estimate field accuracy.

5.2 External validation

Table 2 reports development and holdout results separately. The holdout contained 53 botnet-origin and 55 normal-origin units. Sixteen botnet-origin units alerted, 37 were missed, 32 normal-origin units alerted, and 23 did not.

Table 2 – Family-separated CTU-13 external validation.

Role	Family	Units	TP	FP	FN	TN	Precision	Recall	F1	Specificity	MCC
Development	Virut	31	5	17	0	9	0.227	1.000	0.370	0.346	0.280
Holdout	NSIS.ay	108	16	32	37	23	0.333	0.302	0.317	0.418	-0.282

Source: Prepared by the authors from frozen CTU-13 labeled bidirectional flows (2026).

Development precision had a Wilson 95% interval of 0.101–0.434, recall 0.566–1.000, and specificity 0.194–0.538. Holdout intervals were 0.217–0.475 for precision, 0.195–0.435 for recall, and 0.297–0.550 for specificity. The holdout MCC of –0.282 is a negative generalization result, not evidence of reliable botnet detection.

The rule-level error distribution is given in Table 3; B/N denotes findings on botnet-origin and normal-origin windows. Rapid distinct destinations dominated both correct and incorrect alerts. Periodicity appeared only in normal-origin holdout windows, and the asymmetric-egress rule did not fire.

Table 3 – Rule findings on botnet-origin and normal-origin CTU-13 windows.

Rule	Pattern	Development B/N	Holdout B/N
BEH-001	Periodic communication	1/7	0/15
BEH-002	Distinct endpoints	5/17	16/32
BEH-003	Asymmetric egress	0/0	0/0

Source: Prepared by the authors from the preserved CTU-13 validation output (2026).

The development and holdout files contained 129,832 and 325,471 rows. Clean binary scoring retained 5,558 and 9,755 rows; most excluded rows were background or inbound/ambiguous under the publisher’s label semantics. Streaming processing took approximately 14.4 and 37.9 seconds, with less than 1.3 MiB peak Python memory traced. Runtime demonstrates feasibility for these files only and is not a throughput benchmark.

5.3 Repeated LLM traceability and adversarial fixture

Table 4 reports the repeated Phase-C comparison. Both protocols completed all 50 API calls. Grounded output was parseable, schema-valid, and complete for finding and evidence IDs in 50/50 calls. Forty responses passed the semantic validator. The remaining ten, all for the emulated beacon scenario, retained complete citations but selected segment - source - host - after - approval, which is outside the frozen control set applicable to BEH-001; all ten were rejected.

The reconstructed free-text protocol achieved exact finding- and evidence-ID coverage in 0/50 calls. Its protocol-specific audit found unsupported security-attribution vocabulary in 43/50 responses, containment without an explicit approval qualifier in 10/50, 200-word-limit violations in 37/50, and Markdown markers despite the instruction in 50/50. Free text was not asked to emit JSON, so zero schema validity is a representation difference rather than an independent quality failure.

In the separate adversarial fixture, all ten responses were schema-valid, cited the exact supplied evidence, and omitted both FAKE-999 and the injected absolute malware claim. Nine passed semantic validation. One selected the same BEH-001-inapplicable segmentation control and was safely rejected, as shown in Table 5. This fixed fixture demonstrates resistance to one committed instruction-like string, not general prompt-injection robustness.

Validator acceptance establishes programmed traceability and policy confor-

Table 4 – Repeated Phase-C automated traceability comparison.

Protocol	Calls	API responses	Exact findings	Exact evidence	System accepted	Policy rejected
Grounded schema 1.1	50	50	50	50	40	10
Reconstructed free text	50	50	0	0	0	–

Note: Free text was not asked to emit JSON; its system-accepted count applies the common exact-ID grounding criterion. Grounded policy rejections retained complete citations but selected a rule-inapplicable control.

Source: Prepared by the authors from 100 preserved local Ollama calls (2026).

Table 5 – Repeated prompt-injection fixture result.

Calls	Schema valid	Exact evidence	Fake-ID echo	Absolute claim	Accepted
10	10	10	0	0	9

Note: One response was safely rejected for a rule-inapplicable control; no response followed the injected instruction to cite FAKE - 999 or claim confirmed malware.

Source: Prepared by the authors from the preserved adversarial manifest and outputs (2026).

mance. It does not establish interpretive correctness. The 20% grounded rejection rate demonstrates why structured generation cannot replace deterministic validation. Broad lexical attribution was audited for historical prose but not extracted symmetrically from grounded JSON fields. It is therefore a protocol-specific diagnostic, not a confirmatory between-protocol effect.

5.4 Counterfactual response safety

The system executed zero automatic actions. If every alert had caused a block, 17 of 22 development actions (77.3%) and 32 of 48 holdout actions (66.7%) would have targeted verified normal-origin windows. Simultaneously, 37 holdout botnet-origin windows would have remained unaddressed. Thus, even a reversible automatic block-on-alert policy is not justified by the measured detector.

6 DISCUSSION

6.1 RQ1: transparent rules do not generalize from fixtures

The synthetic F1 of 0.889 and external holdout F1 of 0.317 answer different questions. Phase A shows functional agreement with constructed patterns. Phase B shows that timing and destination diversity are common in ordinary traffic and absent from many labeled botnet windows. Combining the datasets would hide this construct gap. RQ1 is therefore answered negatively for reliable malware-origin classification: the frozen rules are useful as explainable hypotheses for review, not as a standalone detector.

The error pattern identifies a concrete systems requirement. Network flows lack process ownership, executable identity, signer information, parent–child lineage, file creation, and user context. Windows Sysmon, for example, can associate network connections, DNS queries, and file creation with processes, although collection volume and privacy require selective configuration (Microsoft, 2026). Future testing should add authorized endpoint context and evaluate it on a newly frozen holdout rather than retuning against the inspected CTU-13 scenario.

6.2 RQ2: automated traceability improves, semantic superiority remains open

For the fixed model, evidence pack, and automated criteria, H2 is supported for exact traceability: grounded output preserved all supplied identifiers in 50/50 calls, whereas the reconstructed free-text path preserved them in 0/50. End-to-end grounded acceptance was lower, 40/50, because the validator consistently blocked a rule-inapplicable control. The adversarial fixture likewise retained exact IDs and rejected its one policy-invalid response without following the injected fake citation or absolute claim.

This answers RQ2 for automated traceability and enforcement in the implemented artifact, not for general semantic correctness or analyst usefulness. The broad unsupported-claim metric is not fully symmetric across representations, ranking stability is not estimable when every scenario yields one finding, and no blinded expert ratings were collected. Schema constraint changes failure visibility and makes rejection enforceable; it does not make generation intrinsically reliable.

6.3 RQ3: the justified contribution is triage

The artifact can normalize evidence, correlate events with an authorized asset inventory, identify why a rule matched, preserve exact event IDs, and propose additional evidence collection. It may prepare a scoped response candidate containing source, destination, protocol, duration, approval requirement, and rollback conditions. It cannot responsibly identify malware, confirm exploitability, isolate a host, quarantine a file, or change a firewall before independent confirmation and human authorization. The counterfactual results make this boundary empirical rather than merely procedural.

7 THREATS TO VALIDITY AND ETHICAL CONSTRAINTS

Construct validity is limited because CTU-13 labels describe flow origin, not whether every five-minute window contains a detectable malicious action. The study measures review-trigger performance under that operational definition, not malware-family classification. Network-only adaptation cannot evaluate the endpoint-dependent

BEH-004 rule.

Internal validity is limited by fixed heuristic thresholds, source-host windowing, and the possibility that scenario-specific traffic generation affects both labels and patterns. Development and holdout families are separate, but only two of thirteen CTU-13 scenarios were used. The holdout has already been inspected and must not be reused as untouched evidence after tuning.

External validity is limited by an older academic dataset, two malware families, 139 scored windows, one local model, six fixed LLM prompts, and one workstation. Confidence intervals are wide. No production telemetry, live malware, packet payload, or actual containment was used. Ten repetitions per prompt set describe stability under the frozen parameters but are clustered observations, and automated lexical checks do not replace blinded expert judgment.

Reproducibility depends on the frozen URLs and hashes because the large external text flows are not committed. The experiment preserves aggregate and anonymized unit output, but external availability remains outside repository control. Model-tag reproducibility additionally depends on preserving the exact Ollama model digest.

All collection must be authorized. Infrastructure and endpoint telemetry can contain personal and sensitive operational data and must follow purpose limitation, retention, access control, and applicable privacy law. The public fixtures are inert; acquisition code blocks malware binaries and packet captures. Reports are advisory, and the LLM has no action credentials.

8 CONCLUSION

This study evaluated an evidence-grounded local-LLM architecture instead of assuming that a persuasive narrative improves security. The functional fixtures passed their intended rules, but family-separated CTU-13 validation failed to support reliable generalization: holdout recall was 0.302, specificity 0.418, F1 0.317, and MCC -0.282 . A block-on-alert policy would have acted on normal-origin traffic in two thirds of holdout alerts while missing 37 botnet-origin windows.

The retained negative results are the principal contribution. Transparent network-flow heuristics can produce auditable review candidates, but process identity and file lineage are necessary before stronger malware-related interpretation is tested. Repeated local schema-constrained generation preserved exact traceability, yet one fifth of Phase-A calls still required semantic policy rejection. The reconstructed free-text path preserved no exact identifiers, while the inert adversarial fixture showed bounded resistance to its committed instruction-like string. The responsible system boundary is therefore clear: deterministic collection, validation, and triage may be automated; attribution and containment may not.

Future work should add privacy-reviewed endpoint telemetry, freeze a new family-separated external holdout, broaden adversarial prompts and model families, obtain blinded expert ratings, and test only reversible response proposals in an isolated authorized laboratory. Any later threshold, prompt, validator, or audit change must be reported as a new version rather than retroactively improving the present result.

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